

Dataset and Visualization

1. **Create a small synthetic dataset** containing at least 10 two-dimensional points. Use NumPy to store the data.
2. **Visualize the dataset** using a scatter plot. Label the axes and give the plot an appropriate title.

Elbow Method

3. **Implement the Elbow Method for K-Means clustering** on your dataset. Compute the Within-Cluster Sum of Squares (WCSS) for K values ranging from 1 to 5.
4. **Plot the WCSS values** against K and label the axes. Explain how you would use the plot to select the optimal number of clusters.

K-Means Clustering

5. **Perform K-Means clustering** on your dataset using 3 clusters. Assign cluster labels to each data point.
6. **Plot the K-Means clusters** on a scatter plot and show the cluster centroids. Use different colors for each cluster.

Hierarchical Clustering

7. **Compute hierarchical clustering linkage** for your dataset using the Ward method.
8. **Plot a dendrogram** to visualize hierarchical clustering. Label the axes and provide a title.
9. **Assign clusters from the dendrogram** by cutting it to form 3 clusters. Visualize the clusters in a scatter plot.

DBSCAN Clustering

10. **Perform DBSCAN clustering** on your dataset. Choose appropriate values for `eps` and `min_samples`.
11. **Visualize DBSCAN clusters** on a scatter plot. Make sure to indicate noise points if any.

Observations and Comparison

12. **Print observations comparing K-Means, Hierarchical, and DBSCAN clustering** methods. Highlight strengths, limitations, and use cases for each.